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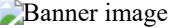
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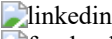
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The 5G Core Network Demystified

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In the [first](#) blog of this 5G Core series, we looked at the concept of cloud-native design, its applications in the 5G network, the benefits and how [Dell and Red Hat](#) are simplifying the deployment and management of cloud-native 5G networks.

With this second blog post we aim to demystify the 5G Core network, its architecture, and how it stands apart from its predecessors. We will delve into the core network functions, the role of Cloud-Native architecture, the concept of network slicing, and how these elements come together to define the 5G Network Architecture.

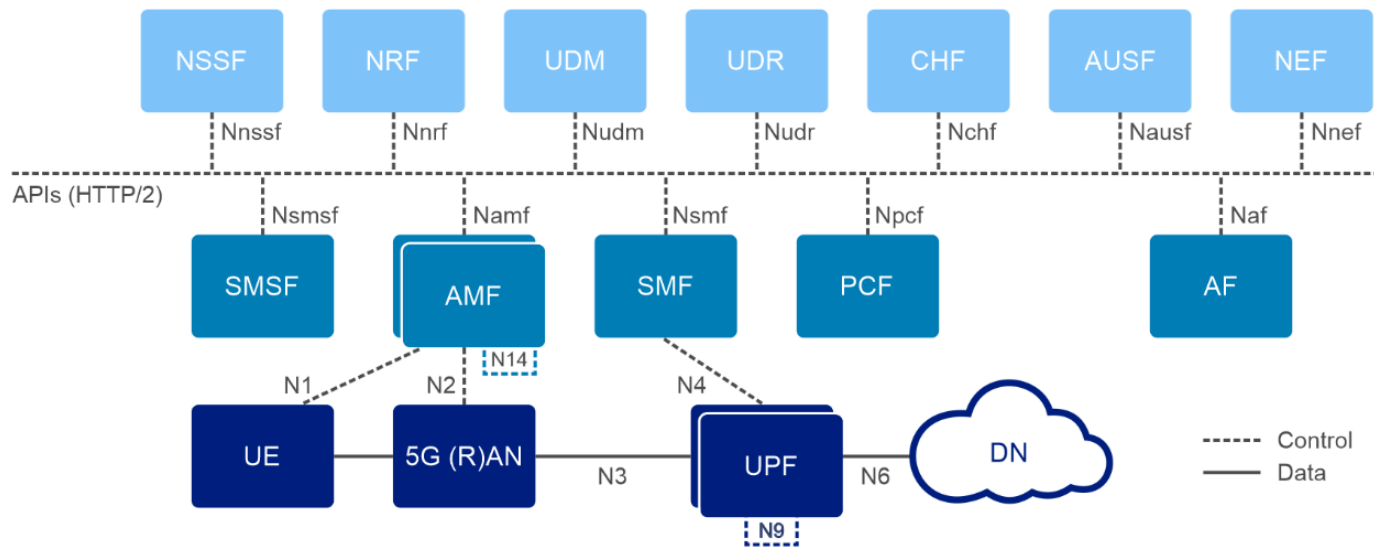
The essence of 5G Core

5G Core, often abbreviated as 5GC, is the heart of the 5G network. It is the control center that governs all the protocols, network interfaces, and services that make the 5G system function seamlessly. The 5G Core is the brainchild of 3GPP (3rd Generation Partnership Project), a standards organization whose specifications cover cellular telecommunications technologies, including radio access, core network and service capabilities, which provide a complete system description for mobile telecommunications.

The 5G Core is not just an upgrade from the 4G core network, it is a radical transformation designed to revolutionize the mobile network landscape. It is built to handle a broader audience, extending its reach to all industry sectors and time-critical applications, such as autonomous driving. The 5G core is responsible for managing a wide variety of functions within the mobile network that make it possible for users to communicate. These functions include mobility management, authentication, authorization, data management, policy management, and quality of service (QOS) for end users.

5G Network Architecture: What You Need to Know

5G was built from the ground up, with network functions divided by service. As a result, this architecture¹ is also known as the 5G core Service-Based Architecture (SBA), The 5G core is a network of interconnected services, as illustrated in the figure below.



3GPP defines that 5G Core Network as a decomposed network architecture with a service-based Architecture (SBA) where each 5G Network Function (NF) can subscribe to and register for services from other NF, using HTTP/2 as a baseline communication protocol.

A second concept in the architecture of 5G is to decrease dependencies between the Access Network (AN) and the Core Network (CN) by employing a unified access-agnostic core network with a common interface between the Access Network and Core Network that integrates diverse 3GPP and non-3GPP access type.

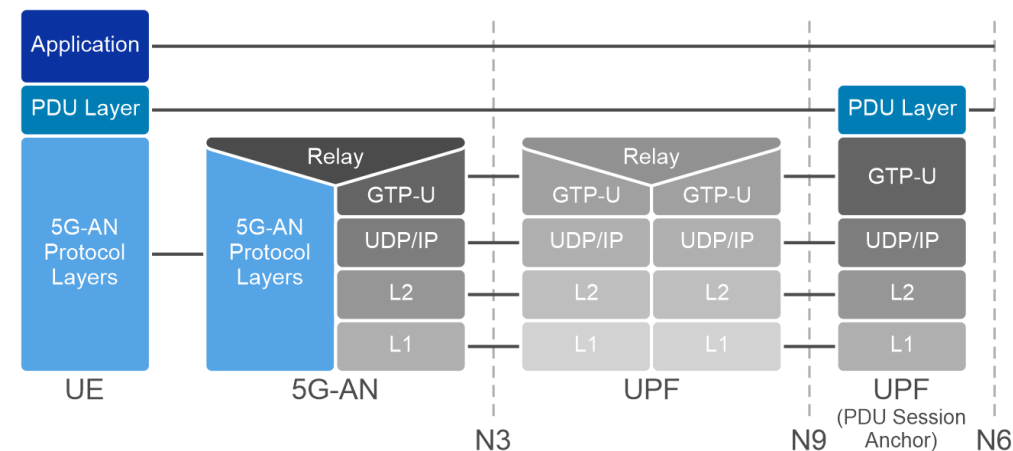
In addition, the 5G core decouples the user plane (UP) (or data plane) from the control plane (CP). This function, which is known as CUPS² (Control & User Plane Separation), was first introduced in 3GPP release 14. An important characteristic of this function being that, in case of a traffic peak, you can dynamically scale the CP functions without affecting the user plane operations, allowing deployment of UP functions (UPF) closer to the RAN and User Equipment (UE) to support use cases like Ultra Reliable low latency Communication (URLLC) and achieve benefits in both Capex and Opex.

5G Core Network Functions and What They Do

The 5G Core Network is composed of various network functions, each serving a unique purpose. These functions communicate internally and externally over well-defined standard interfaces, making the 5G network highly flexible and agile. Let's take a closer look at some³ of the critical 5G Core Network functions:

User Plane Function (UPF)

The User Plane Function is a critical component of the 5G core network architecture. It oversees the management of user data during the data transmission process. The UPF serves as a connection point between the RAN and the data network. It takes user data from the RAN and performs a variety of functions like as packet inspection, traffic routing, packet processing, and QoS enforcement before delivering it to the Data Network or Internet. This function allows the data plane to be shifted closer to the network edge, resulting in faster data rates and shorter latencies. The UPF combines the user traffic transport functions previously performed in 4G by the Serving Gateway (S-GW) and Packet Data Network Gateway (P-GW) in the 4G Evolved Packet Core (EPC).



UPF Interfaces/reference points with employed protocols:

- N3 (GTP-U): Interface between the RAN (gNB) and the UPF
- N9 (GTP-U): Interface between two UPF's (i.e the Intermediate I-UPF and the UPF Session Anchor)

- N6 (GTP-U): Interface between the Data Network (DN) and the UPF
- N4 (PFCP): Interface between the Session Management Function (SMF) and the UPF

Session Management Function (SMF)

The Session Management Function (SMF) is crucial element that make up the 5G Core Network responsible for establishing, maintaining, and terminating network sessions for User Equipment (UE). The SMF carries out these tasks using network protocols such as Packet Forwarding Control Protocol (PFCP) and Network Function-specific Service-based interface (Nsmf).

SMF communicates with other network functions like the Policy Control Function (PCF), Access and Mobility Management Function (AMF), and the UPF to ensure seamless data flow, effective policy enforcement, and efficient use of network resources. It also plays a significant role in handling Quality of Service (QoS) parameters, routing information, and charging characteristics for individual network sessions.

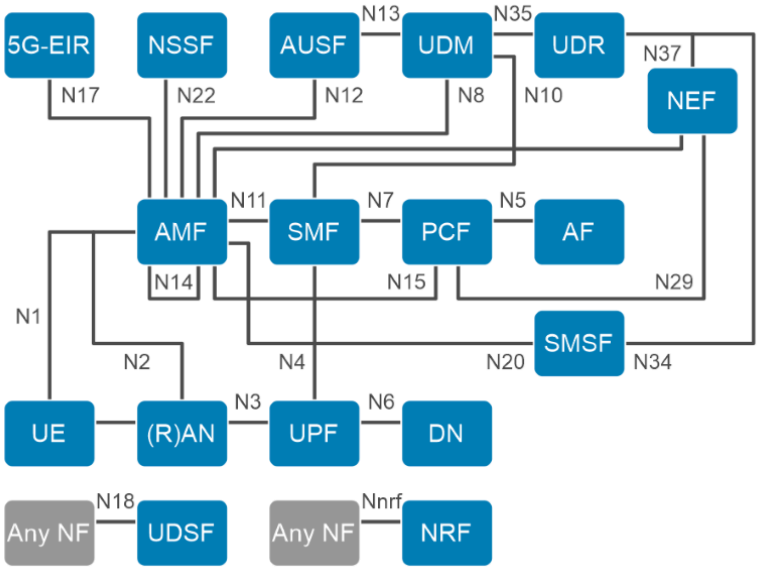
SMF brings some control plane functionality of the serving gateway control plane (SGW-C) and packet gateway control plane (PGW-C) in addition to providing the session management functionality of the 4G Mobility Management Entity (MME).

Access and Mobility Management Function (AMF)

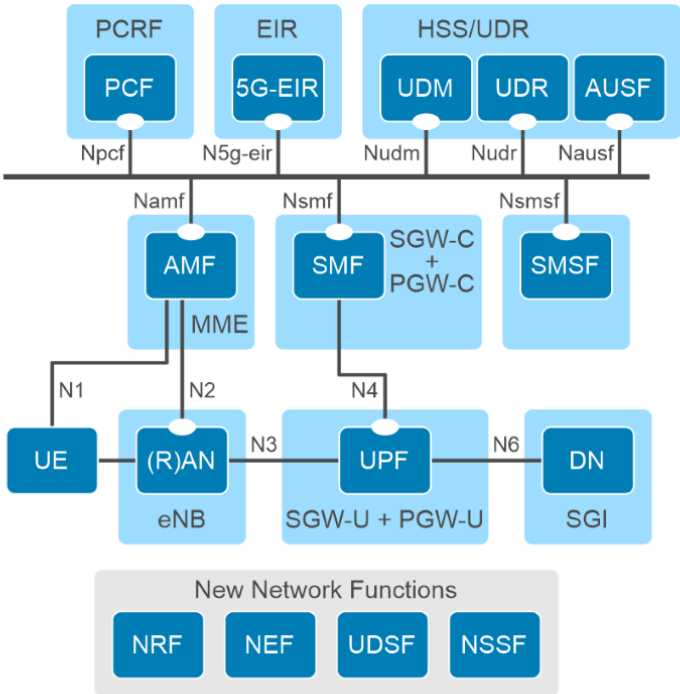
The Access and Mobility Management Function (AMF) oversees the management of connections and mobility. It receives policy control, session-related, and authentication information from the end devices and passes the session information to the PCF, SMF and other network functions. In the 4G/EPC network, the corresponding network element to the AMF is the Mobility Management Entity. While the MME's functionality has been decomposed in the 5G core network, the AMF retains some of these roles, focusing primarily on connection and mobility management, and forwarding session management messages to the SMF.

Additionally, the AMF retrieves subscription information and supports short message service (SMS). It identifies a network slice using the Single Network Slice Selection Assistance Information (S- NSSAI), which includes the Slice/Service Type (SST) and Slice Differentiator (SD). The AMF's operations enable the management of Registration, Reachability, Connection, and Mobility of UE, making it an essential component of the 5G Core Network.

5G Core Network Functions



Mapping of 5G Core Network Functions to EPC



Policy Control Function (PCF)

The Policy Control Function (PCF) provides the framework for creating policies to be consumed by the other control plane network functions. These policies can include aspects like QoS, Subscriber Spending/Usage Monitoring, network slicing management, and management of subscribers, applications, and network resources. The PCF in the 5G network serves as a policy decision point, like the PCRF (Policy and Charging Rules Function) in 4G/EPC Network. It communicates with other network elements such as the AMF, SMF, and Unified Data Management (UDM) to acquire critical information and make sound policy decisions.

Unified Data Management (UDM) and Unified Data Repository (UDR)

The Unified Data Management (UDM) and Unified Data Repository(UDR) are critical components of the 5G core network. The UDM maintains subscriber data, policies, and other associated information, while the UDR stores this data. They collaborate to conduct data management responsibilities

that were previously handled by the HSS (Home Subscriber Server) in the 4G EPC. When compared to the HSS, the UDM and UDR provide greater flexibility and efficiency, supporting the enhanced capabilities of the 5G network.

Network Exposure Function (NEF)

The Network Exposure Function (NEF) is another key component of 5G core network that enables network operators to securely expose network functionality and interfaces on a granular level by creating a bridge between the 5G core network and external application (E.g., internal exposure/re-exposure, Edge Computing). The NEF also provides a means for the Application Functions (AFs) to securely provide information to 3GPP network (E.g., Expected UE Behavior).

The NEF northbound interface is between the NEF and the AF. It specifies RESTful APIs that allow the AF to access the services and capabilities provided by 3GPP network entities and securely exposed by the NEF. It communicates with each NF through a southbound interface facilitated by a northbound API. The 3GPP interface refers to the southbound interface between NEF and 5G network functions, such as the N29 interface between NEF and Session Management Function (SMF), the N30 interface between NEF and Policy Control Function (PCF), and so on.

By opening the network's capabilities to third-party applications, NEF enables a seamless connection between network capabilities and business requirements, optimizing network resource allocation and enhancing the overall business experience.

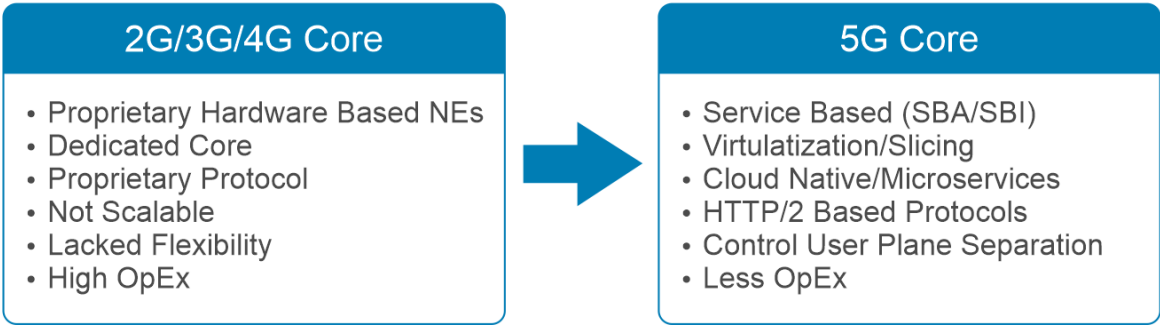
Network Repository Function (NRF)

The Network Resource Function (NRF) serves as critical component required to implement the new service-based architecture in the 5G core network which serves as a centralized repository for all NF's instances. It is in charge of managing the lifecycle of NF profiles, which includes registering new profiles, updating old ones, and deregistering those that are no longer in use. The NRF offers a standards-based API for 5G NF registration and discovery.

Technically, NRF operates by storing data about all Network Function (NF) instances, including their supported functionalities, services, and capacities. When a new NF instance is instantiated, it registers with the NRF, providing all the necessary details. Subsequently, any NF that needs to communicate with another NF can query the NRF for the target NF's instance details. Upon receiving this query, the NRF responds with the most suitable NF instance information based on the requested service and capacity.

How Does the 5G Core Differ from Previous Generations?

The primary architectural distinction between the 5G Core and the 4G EPC is that the 5G Core makes use of the Service-Based Architecture (SBA) with cloud-native flexible configurations of loosely coupled and independent NFs deployed as containerized microservices. The microservices based architecture provides the ability for NFs to scale and upgrade independently of each other which is significant benefit to CSPs. The 4G EPC, on the other hand, employs a flat architecture for efficient data handling with network components deployed as physical network elements in most cases and the interface between core network elements was specified as point-to-point running proprietary protocols and was not scalable.

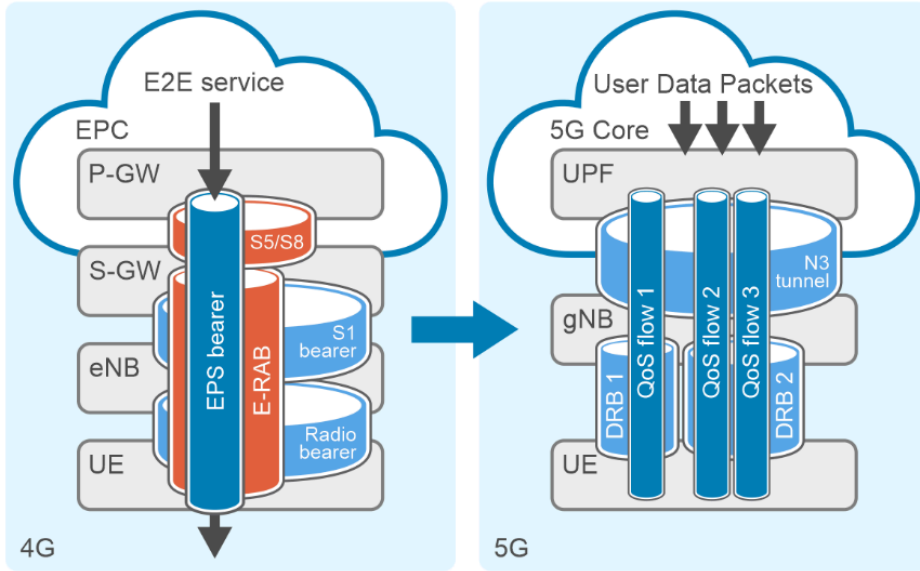


Another significant distinction between 5G Core and EPC is the formation of the control plane (CP). The control plane functionality is more intelligently shared between Access and Mobility Management Functions (AMF) and Session Management Functions (SMF) in the 5G Core than the MME and SGW/PGW in the 4G/EPC. This separation allows for more efficient scaling of network resources and improved network performance.

In addition to the design and functional updates, the business' priorities with 5G have been updated. With 5GC, CSPs are moving away from proprietary, vertically integrated systems and shifting to cloud-native and open source-based platforms like [Red Hat OpenShift Container Platform](#) that runs on industry standard hardware. This helps improve the responsiveness while also cutting the operating expenses will be the primary focus going forward with 5G Core for CSPs.

Key distinctions between the 4G LTE and 5G QoS models

The key distinctions between 4G LTE and 5G QoS models primarily lie in their approach to quality-of-service enforcement and their level of complexity. In 4G LTE, QoS is enforced at the EPS bearer level (S5/S8 + E-RAB) with each bearer assigned an EPS bearer ID. On the other hand, 5G QoS is a more flexible approach that enforces QoS at the QoS flow level. Each QoS flow is identified by a QoS Flow ID (QFI).



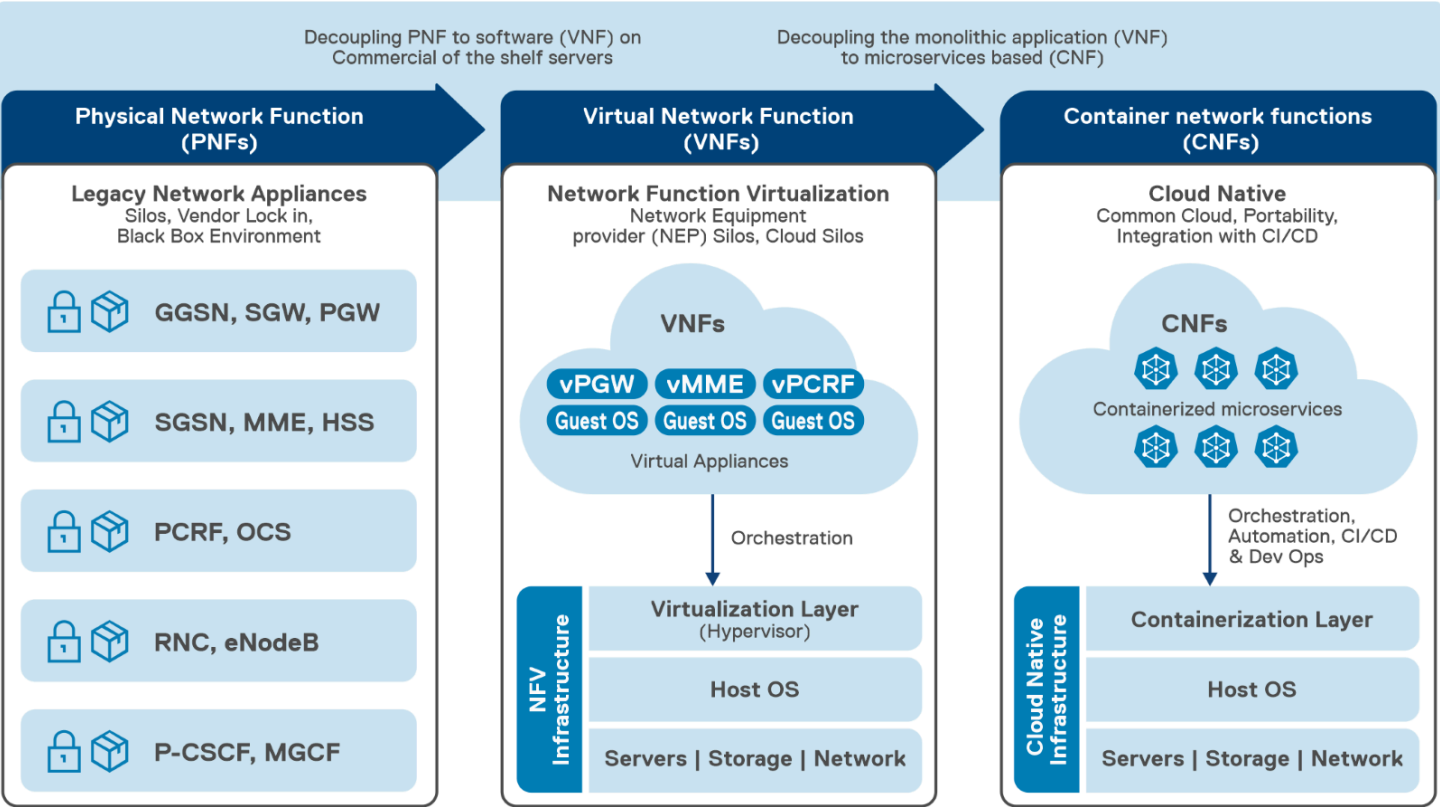
Furthermore, the process of ensuring end-to-end QoS for a Packet Data Unit (PDU) session in 5G involves packet classification, user plane marking, and mapping to radio resources. Data Packets are classified into QoS flows by UPF using Packet Detection Rules (PDRs) for downlink and QoS rules for uplink.

5G leverages Service Data Adaptation Protocol (SDAP) for mapping between a QoS flow from the 5G core network and a data radio bearer (DRB). This level of control and adaptability provides an improved QoS model in 5G as compared to 4G networks.

The Power of Cloud-Native Architecture in 5G Core

One of the standout features of the 5G Core is its cloud-native architecture. This architecture allows the 5G core network to be built with microservices that can be reused for supporting other network functions. The 5G core leverages technologies like microservices, containers, orchestration, CI/CD pipelines, APIs, and service meshes, making it more agile and flexible.

With Cloud-Native architecture, 5G Core can be easily deployed and operated, offering a cost-effective solution that complies with regulatory requirements and supports a wide range of use cases. The adherence to cloud-native principles is of utmost importance as it allows for the independent scaling of components and their dynamic placement based on service demands and resource availability. This architecture also allows for network slicing, which enables the creation of end-to-end virtual networks on top of a shared infrastructure.



Network Slicing: Enabling a Range of 5G Services

Network Slicing is considered as one of the key features by 3GPP in 5G. A network slice can be looked like a logical end-to-end network that can be dynamically created. A UE may access to multiple slices over the same gNB, within a network slice, UEs can create PDU sessions to different Gateways via Data network name (DNNs). This architecture allows operators to provide a custom Quality of Service (QoS) for different services and/or customers with agreed upon Service-level Agreement (SLA).

The Network Slice Selection Function (NSSF) plays a vital role in the network slicing architecture of 5G Core. It facilitates the process of selecting the appropriate network slice for a device based on the Network Slice Selection Assistance Information (NSSAI) specified by the device. When a device sends a registration request, it mentions the NSSAI, thereby indicating its network slice preference. The NSSF uses this information to determine which network slice would best meet the device's requirements and accordingly assigns the device to that network slice. This ability to customize network slices based on specific needs is a defining feature of 5G network slicing, enabling a single physical network infrastructure to cater to a diverse range of services with contrasting QoS requirements. To read and learn more on Network Slicing check out this amazing blog post [To slice or not to slice | Dell Technologies Info Hub](#).

Next steps

To learn how Dell and Red Hat are helping CSPs in their cloud-native journey, see the blog [Cloud-native or Bust: Telco Cloud Platforms and 5G Core Migration](#) on Info Hub. In the next blog of the 5G Core series, we will explore the collaboration between Dell Technologies and Red Hat to simplify operator processes, starting from the initial technology onboarding all the way to Day 2 operations. The focus is on deploying a telco cloud that supports 5G core network functions using Dell Telecom Infrastructure Blocks for Red Hat.

To learn more about about Telecom Infrastructure Blocks for Red Hat, kindly visit our website [Dell Telecom Multi-Cloud Foundation solutions](#).

¹ The 5G Architecture shown here is the simplified version, there are other 5G NFs like UDSF, SCP, BSF, SEPP, NWDAF, N3IWF etc. not shown here.

² CUPS is a pre-5G technology (5G Standalone (SA) was introduced in 3GPP Rel-15). 5G SA offers more innovation with the ability to change anchors (SSC Mode 3), daisy chain UPFs, and connect to multiple UPFs.

³ The Network Functions mentioned in this section are a subset of the standardized NFs in 5G Core network.

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Gaurav Gangwal works in Dell's Telecom Systems Business (TSB) as a Technical Marketing Engineer on the Product Management team. He is currently focused on 5G products and solutions for RAN, Edge, and Core. Prior to joining Dell in July 2022, he worked for AT&T for over ten years and previously with Viavi, Alcatel-Lucent, and Nokia. Gaurav has an Engineering degree in Electronics and Telecommunications and has worked in the telecommunications industry for about 14+ years. He currently resides in Bangalore, India.